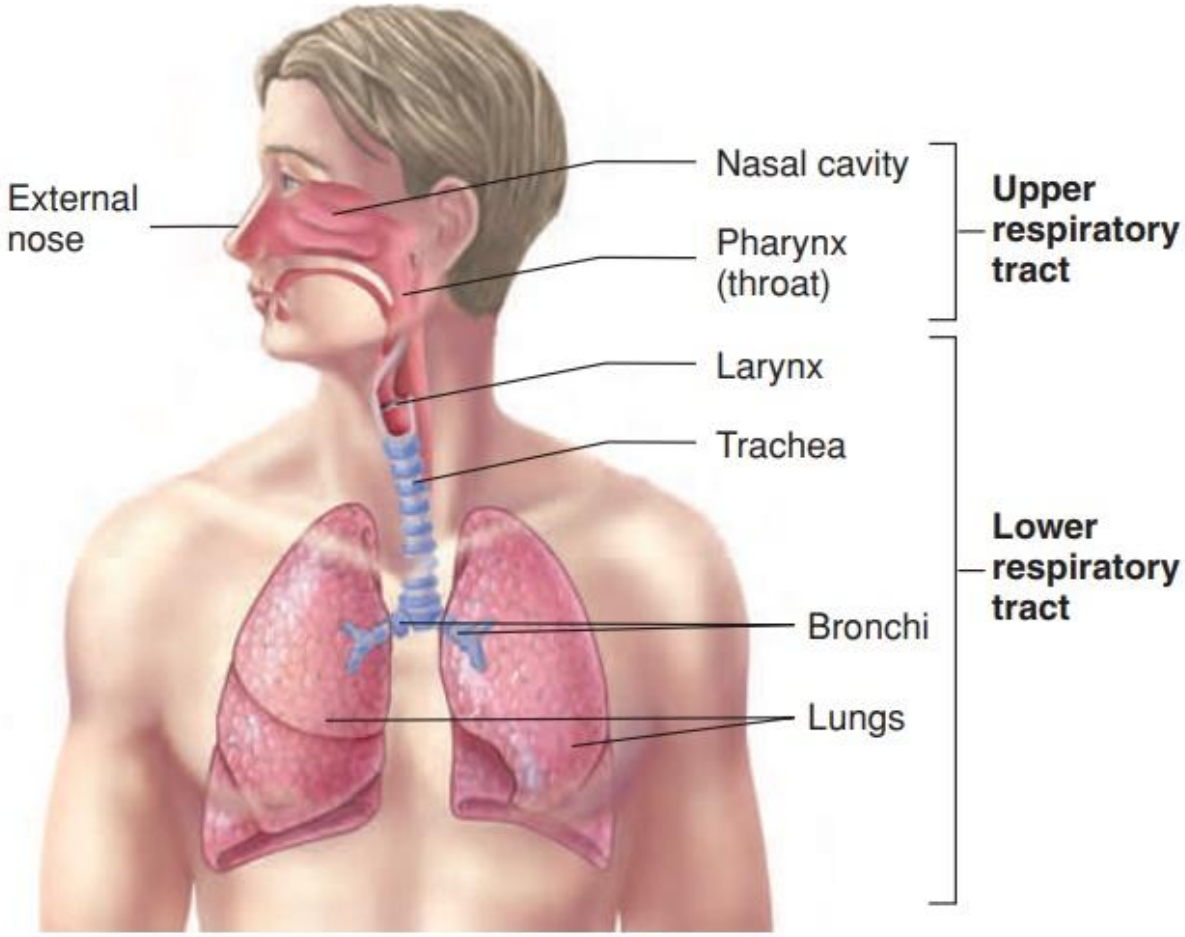
# HISTOLOGY OF LUNG



# THE UPPER & LOWER RESPIRATORY TRACTS



# THE BRONCHIAL TREE

Respiratory system is made up of **nose, pharynx, larynx, trachea, bronchi & lungs**

**Structural divisions**

- **Upper respiratory tract**
  - above the larynx
- **Lower respiratory tract**
  - below the larynx

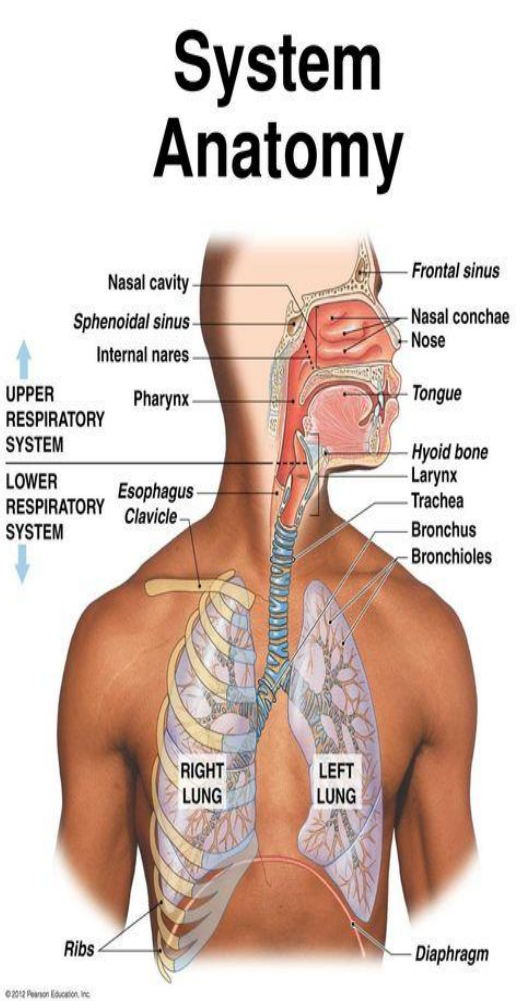
**Functional divisions**

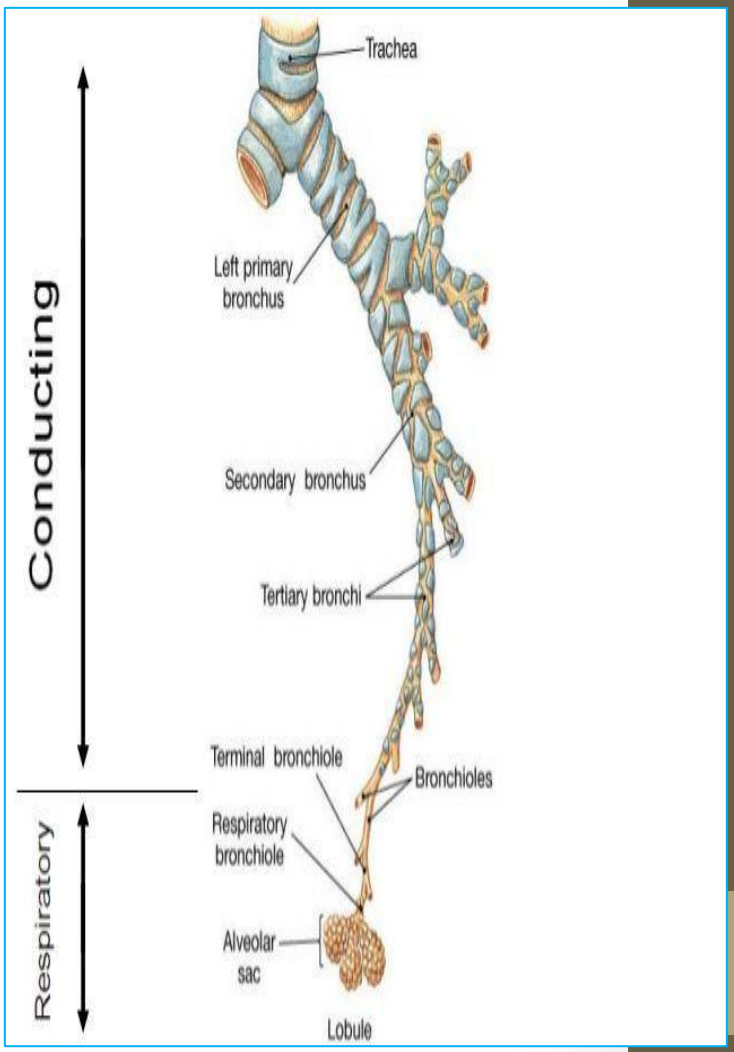
- ***Conducting portion***: cavities and tubes = passageways for air

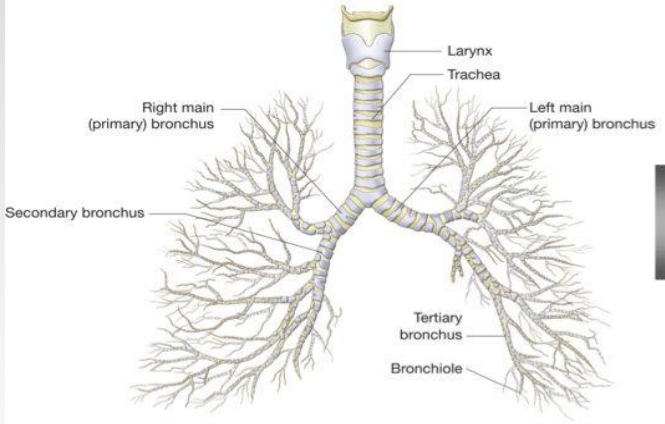
  Nose, pharynx, larynx, trachea, bronchi, bronchioles, and terminal bronchioles.

- ***Respiratory portion***: ***gas exchange:***

  Respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli.






# Bronchial Tree

(Cartilage + smooth muscle)

(Smooth muscle) (no cartilage)

(No cartilage, few/no smooth muscle)

**Bronchi** → **Bronchioles** → **Alveolar structures**

- **Bronchi**
  - Primary
  - Secondary (lobar)
  - Tertiary (segmental)
- **Bronchioles**
  - Intralobular
  - Terminal
  - Respiratory
- **Alveolar structures**
  - Alveolar ducts
  - Alveolar sacs
  - Alveoli

**Trachea**

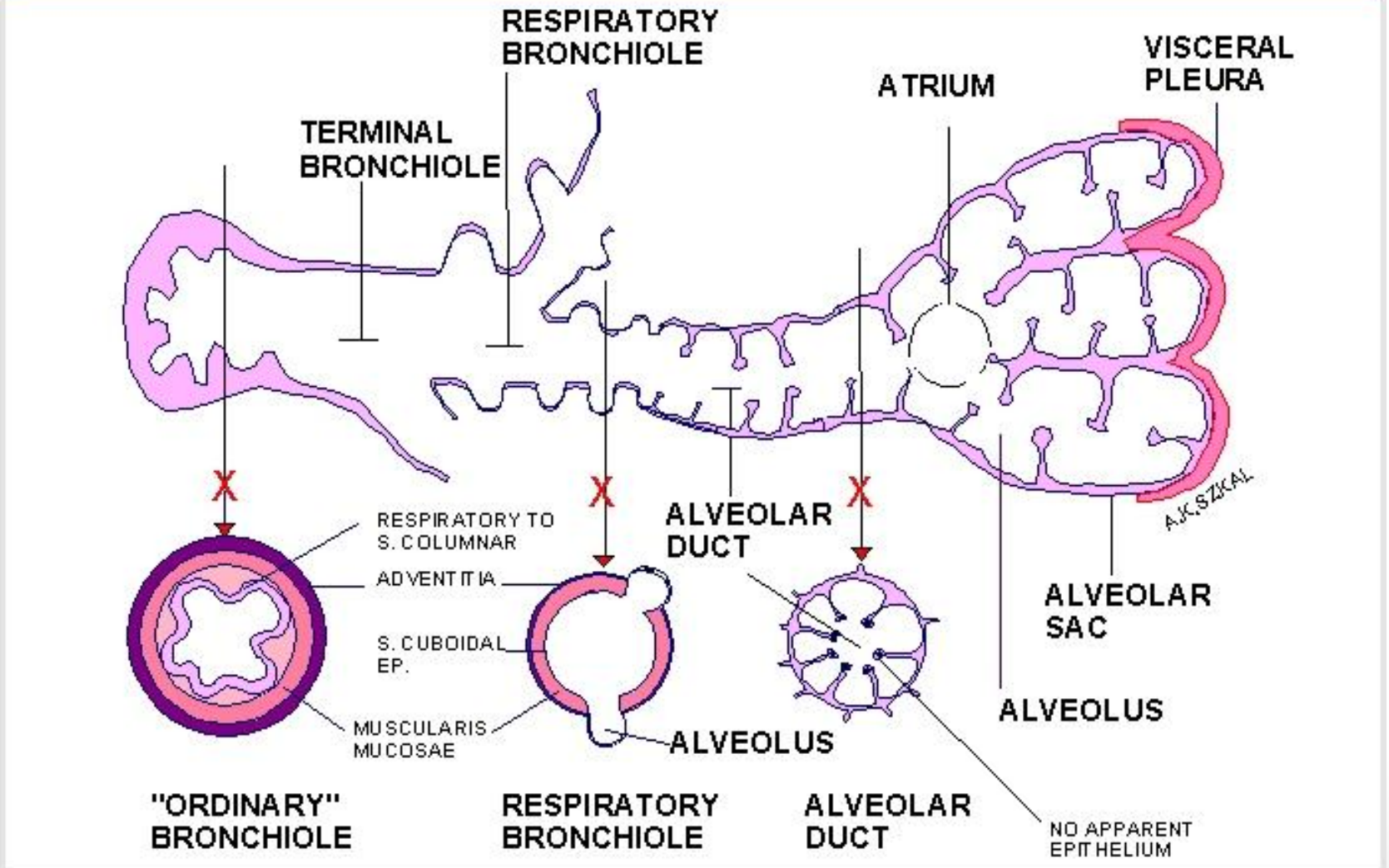
**Which parts here constitute the conducting portion of the respiratory tree?**

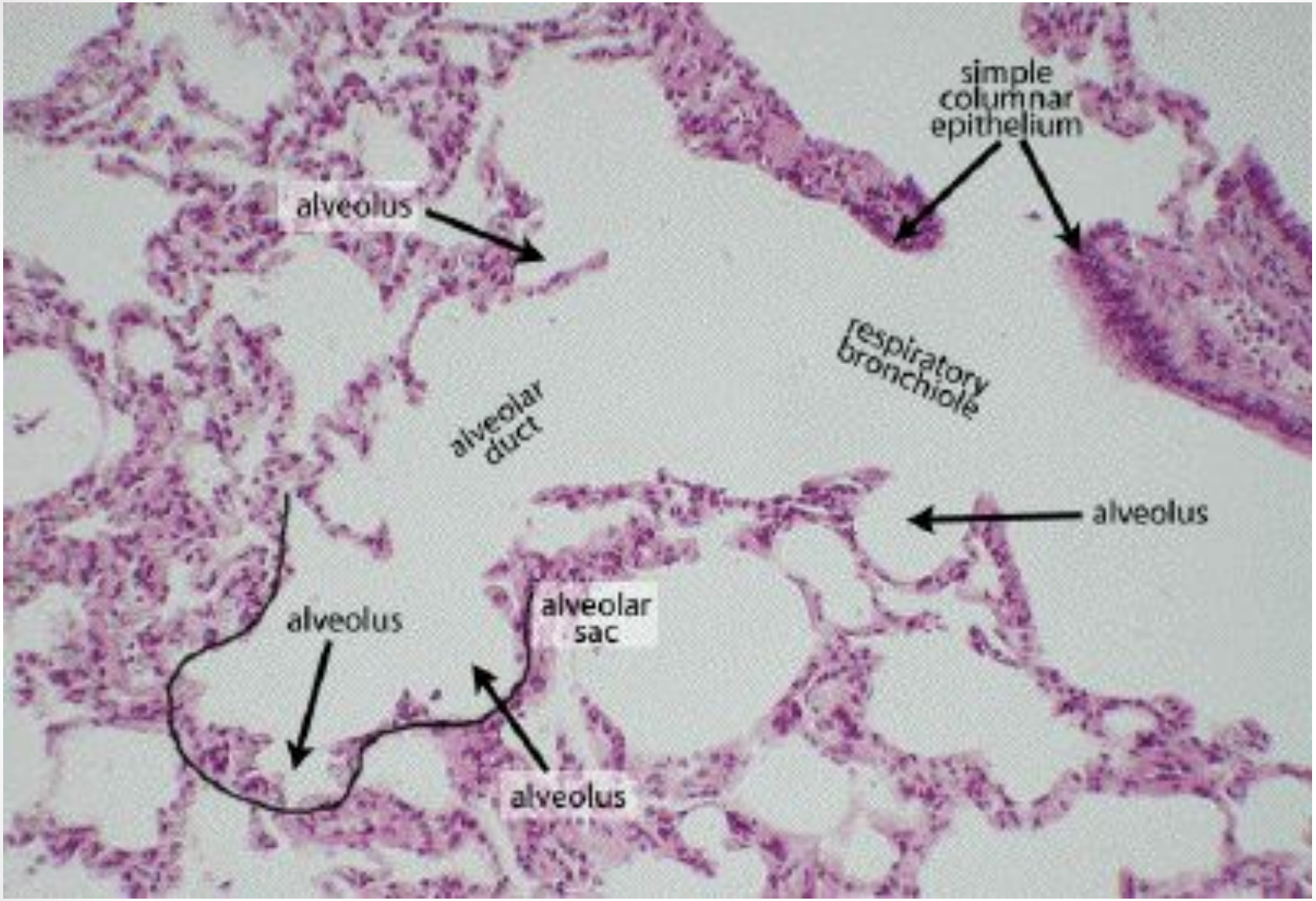
**Know this chart**

# Features of airways within the lungs

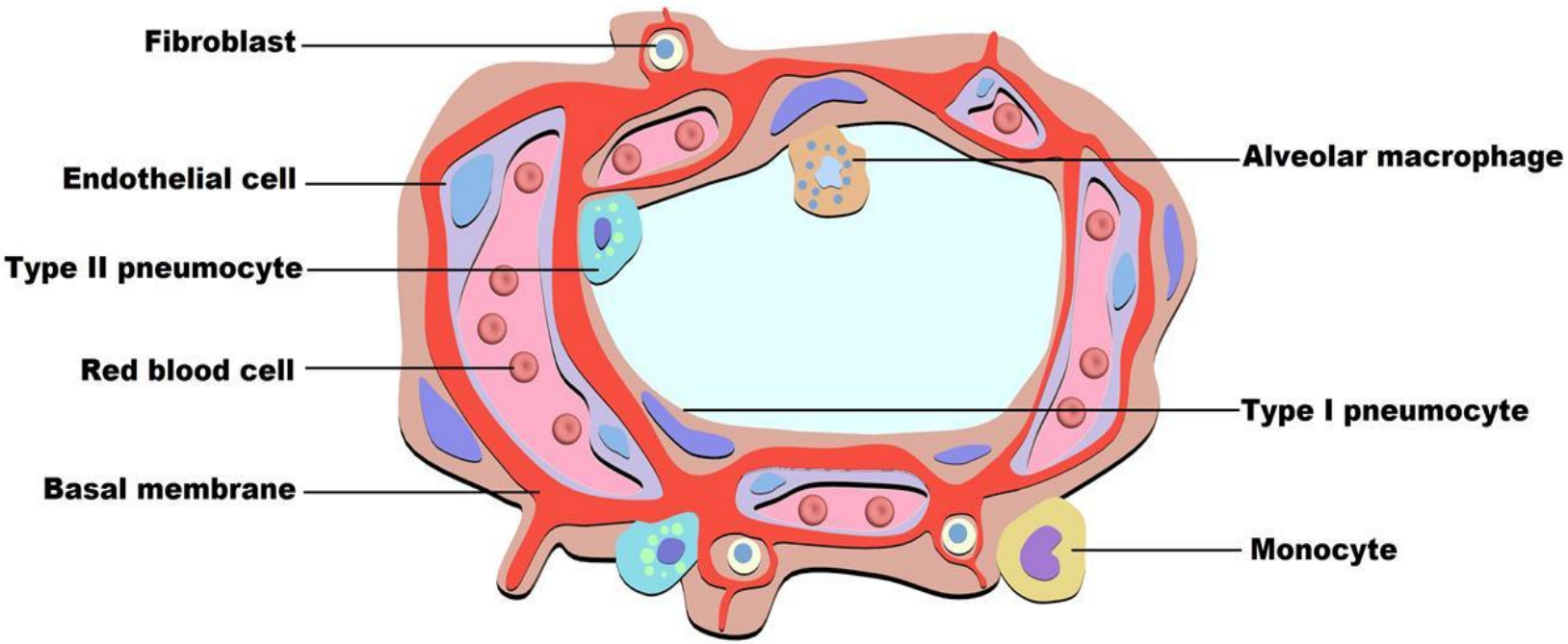
| Airway region | Epithelium | Components in wall | Functions |
|---|---|---|---|
| **Bronchi** | Respiratory | Hyaline cartilage plates and smooth muscle | Conduct air |
| **Bronchioles** | Simple cuboidal to columnar with cilia and club cells | First region lacking cartilage plates, still contains smooth muscle | Conduct air and undergo constriction and dilation |
| **Terminal bronchioles** | Simple cuboidal with cilia and club cells | Smooth muscle, but layers are thinner | Conduct air and produce secretions that protect the airways and function as surfactant |
| **Respiratory bronchioles** | Simple cuboidal with cilia and club cells, regions with alveolar epithelium | Small bundles of smooth muscle | Conduct air, produce secretions, and participate in some gas exchange |
| **Alveolar ducts** | Areas of simple cuboidal interspersed with alveolar epithelium | Small, thin bundles of smooth muscle | Conduct air and participate in gas exchange |
| **Alveoli** | Type I and II pneumocytes | Reticular and elastic fibers | Main site for gas exchange, surfactant production |

# HISTOLOGY OF BRONCHIOLES




simple
columnar
epithelium
alveolus
respiratory
bronchiole
alveolar
duct
alveolus
alveolus
alveolar
sac
alveolus

# Pulmonary Alveoli

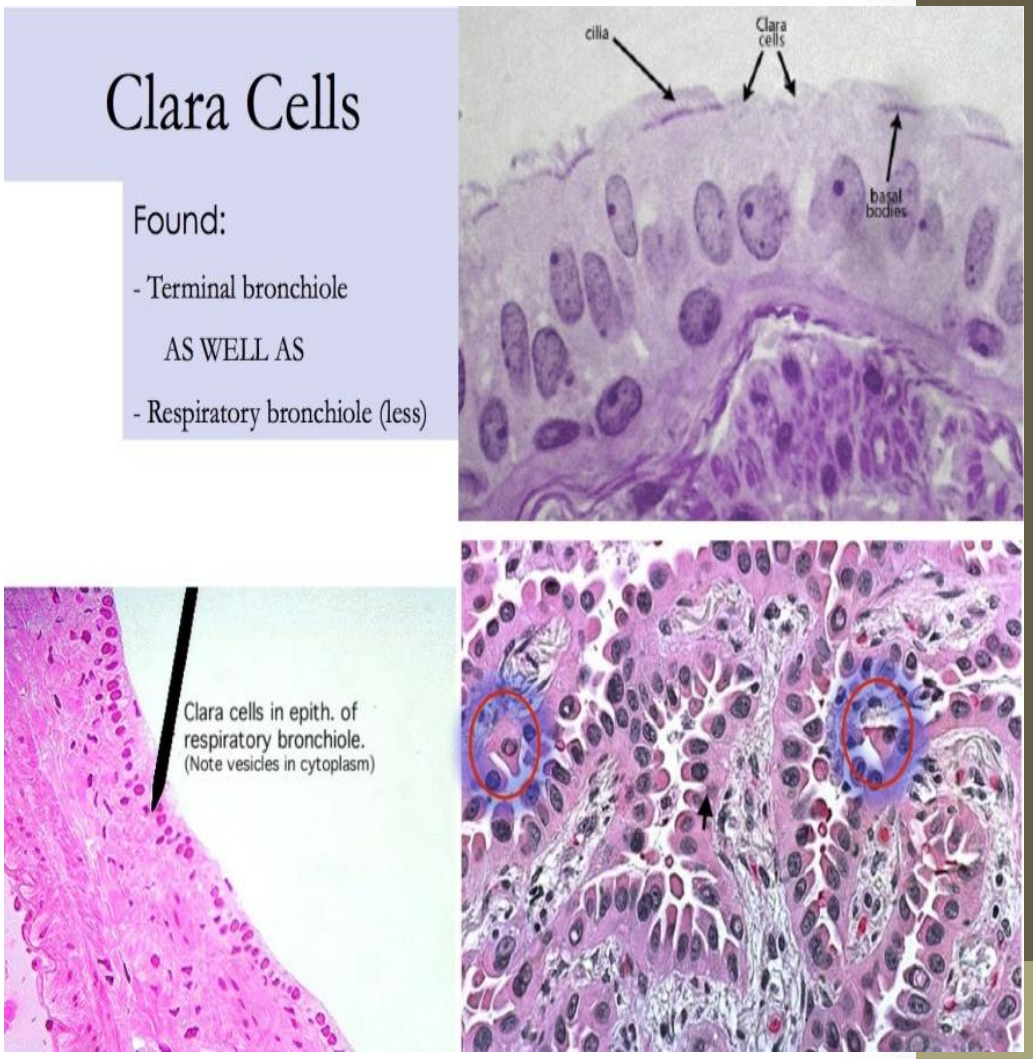


# Clara cells (club cells)

- Numerous in the cuboidal epithelium of terminal bronchioles
- Dome-shaped, non-ciliated & containing secretory granules
- Have abundant RER, Golgi apparatus, mitochondria & some SER

**Functions:**

- Secretion of surfactant
- Detoxification
- Secretion of antimicrobial peptides
- Serve as stem cells

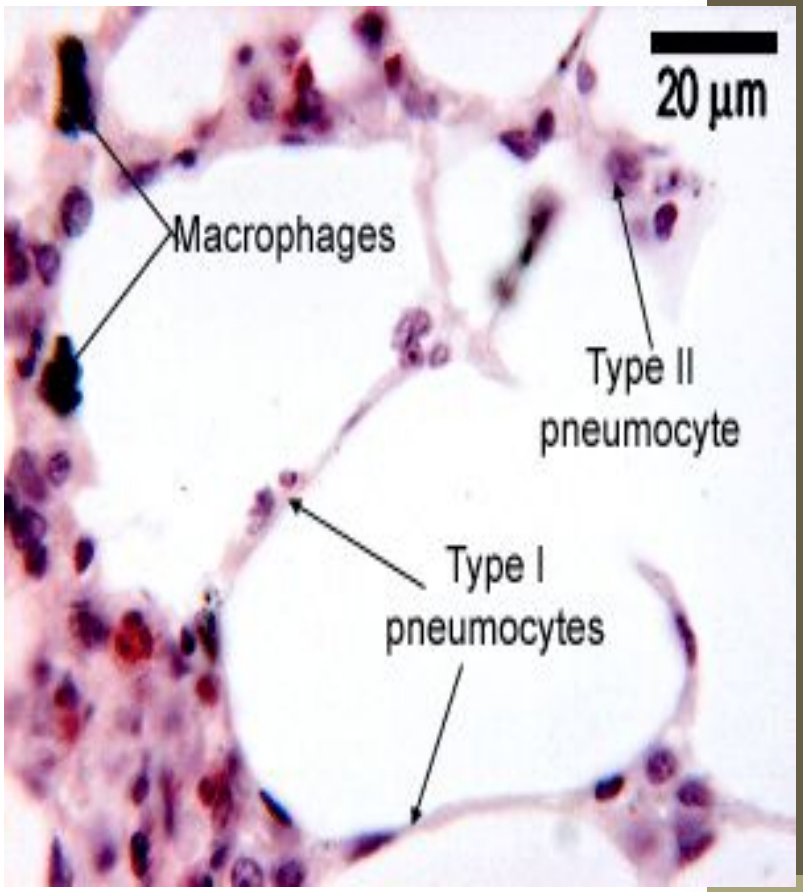


# Type 1 pneumocytes

- Extremely thin flat squamous cells line almost 95% of each alveolus
- Joined by occluding junctions
- Cannot divide mitotically

**Function:**

- Main function is to provide barrier of minimal thickness that allows gaseous exchange between air and blood

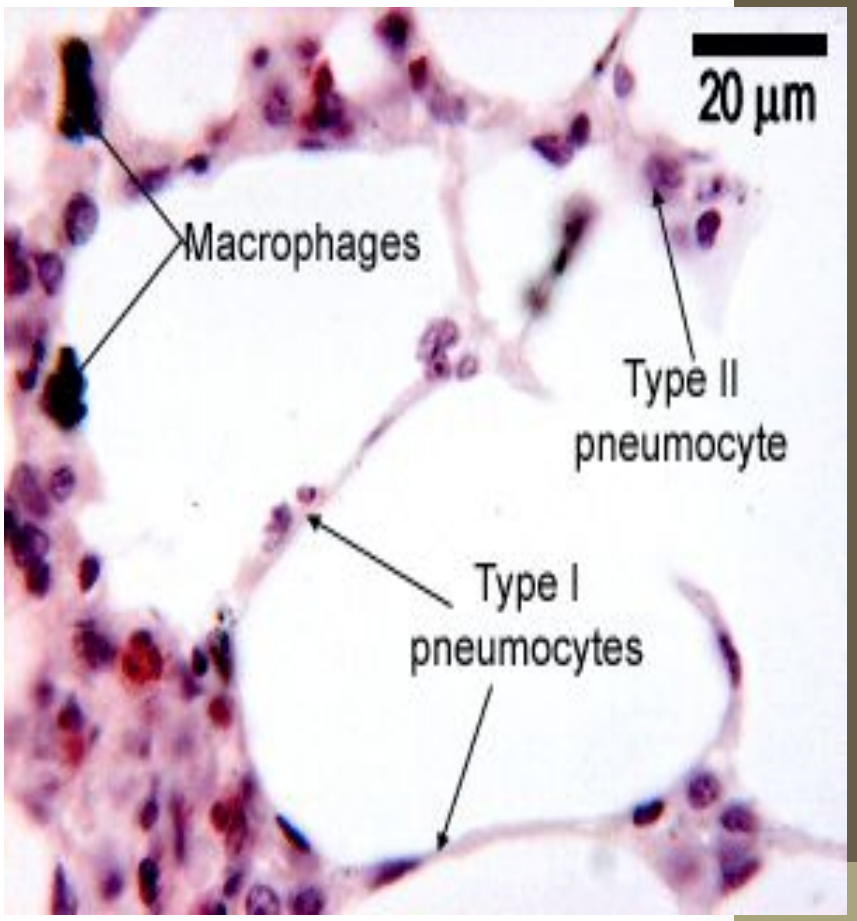


# Type 2 pneumocytes

- Cuboidal cells interspersed among type 1 pneumocytes
- Linked together by occluding junctions & desmosomes
- Divide mitotically to replace their own population & also provide progenitor cells for type 1 population

**Function:**

- Secretion of surfactant

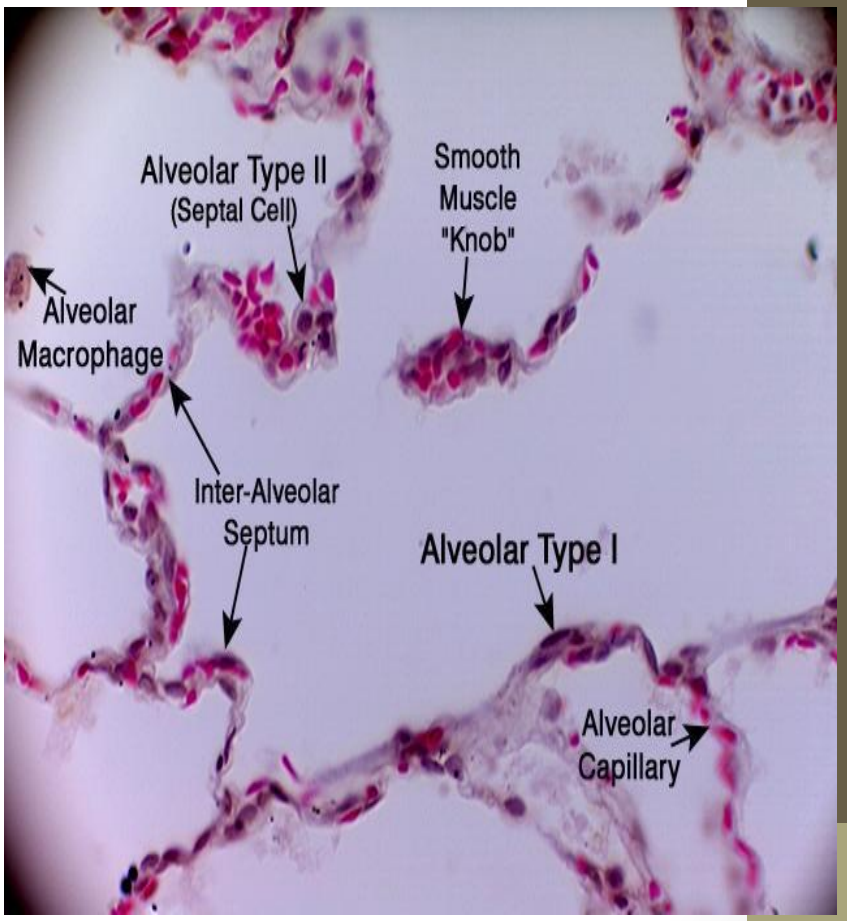


# Alveolar macrophages (dust cells)

- Derived from blood monocytes & belong to mononuclear phagocyte system
- Reside in the lumen of pulmonary alveoli & inter-alveolar septa
- Slightly darker due to their content of dust & carbon from air hence named ‘’dust cells’’

**Function:**

- Phagocytosis of inhaled particulate matter like carbon, pollen, dust particles & bacteria



# Thank you!